



Vehicle classification with single multi-functional magnetic sensor and optimal MNS-based CART



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ABSTRACT

To take advantages of magnetic sensor technology in terms of cost, size, weight, power consumption and wireless communication, a wireless multi-functional magnetic sensor was designed and developed. Then, a novel method with single multi-functional magnetic sensor and optimal Minimum Number of Split-sample (MNS)-based Classification and Regression Tree (CART) algorithm was proposed in this paper to classify on-road vehicles. The sensor was deployed on the road to acquire real-time vehicle waveform data. The decision tree model based on CART algorithm was used to execute on-line vehicle classification in the sensor node. Eight speed-independent time-domain waveform features were extracted as the model inputs. This paper trained the decision tree model by using vehicle samples derived from the multi-functional magnetic sensor and pruned the optimal decision tree with a Minimum Error Pruning (MEP) rule to obtain an optimal pruning tree which is more robust to new samples. Some experiments were implemented by different sample sets and classification methods. The results showed that the proposed method achieved on-line vehicle classification in the sensor node. For the field sample sets with two vehicle classes, the average accuracy rates of test samples were 88.9% and 94.4% in the original samples and swapping samples respectively. Besides higher accuracy, the method also has a better sample robustness, which is easy to classify new samples. The comparison results of current methods also showed that the proposed method has some advantages in aspects of accuracy rate, sample robustness and execution time.

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1. Introduction

In recent years, Intelligent Transportation System (ITS) programs have been initiated in many countries [1]. As a part of ITS, on-line vehicle classification system plays an important role since it gathers some critical parts of the data used in ITS.

Over the past few years, vehicle classification has been widely studied and become an important source of

information for transportation design and management. A vehicle classification system is essential for traffic management, automatic toll collection, parking optimization, law enforcement, autonomous navigation, etc. [2]. Moreover, transportation agencies collect vehicle classification information to create highway maintenance plan, evaluate highway usage and optimize the deployment of various resources on the road [3]. Electronic toll collection and automatic enforcement also require accurate identification of vehicle classes on highways so that vehicles of different classes are charged accordingly [4,5]. In traffic safety research and implementation, the

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distribution of trucks in traffic is a critical design factor, due to its significantly lower speed and larger size [6]. In addition, vehicle classification information allows us to analyze crash data with certain vehicle classes, which support the derivation of more effective countermeasure for the safety enhancement. In pavement design and management, pavement life is estimated according to the distribution of vehicle types running over it, and this distribution may be used to schedule re-surfacing. Furthermore, the distribution of vehicle types also provides valuable data to the prediction of highways capacity, assessment of the effectiveness of traffic legislation, weight enforcement strategies and environmental impact studies [6]. Hence, these needs have made engineers and researchers keep developing traffic surveillance systems capable of providing accurate and reliable vehicle classification information [4].

Currently, various technologies are used for vehicle classification. Loop [7,8], video [2,9,10], acoustic [11,12], and other sensors [13] are used in vehicle classification systems, and a large number of researchers devote themselves to the models and algorithms for vehicle classification by those sensors. Jeng et al. [8] proposed a two-step algorithm to execute vehicle classification using the inductive loop signature technology. Employing some speed estimation advances, Coifman and Kim [7] brought length based vehicle classification to single-loop detectors. Then their classification methodology was evaluated against concurrent measurements from video and dual-loop detectors. For video and acoustic sensors, Kafai and Bhanu [2] presented a stochastic multiclass vehicle classification system which classifies a vehicle into one of four classes: sedan, pickup truck, SUV/minivan, and unknown. A feature set of tail light and vehicle dimensions was extracted which fed a feature selection algorithm to define a low-dimensional feature vector. To obtain more complete knowledge regarding vehicles' acoustic characteristics, Guo et al. [11] proposed a fusion approach to combine two sets of features, in which various aspects of an acoustic signature were emphasized individually. Next, they proposed a modified Bayesian based fusion method to take advantage of matching each specific feature set with its favored classifier. Thus, feature-based methods have been commonly used for vehicle classification.

However, existing technologies such as piezoelectric sensors, inductive loops and video imaging have very high installation and maintenance costs. Some technologies such as infrared sensors, video imaging are sensitive to traffic, environmental, and weather conditions [3]. Furthermore, the performance of microwave radar, infrared sensor and video imaging is often affected by Doppler effects or noise introduced by moving vehicles and terrain variations. In contrast, magnetic sensors are less sensitive to weather, terrain, noise and Doppler effects. Moreover, magnetic sensors have advantages in terms of cost, size, weight, power consumption and wireless communication [14]. Hence, applying magnetic sensors to vehicle classification is a promising research topic.

Recently, some methods on vehicle classification or re-identification utilizing magnetic sensors have been reported in literature [1,3,5,6,15]. In order to improve the recognition rate by a single magnetic sensor, Lan et al. [1] proposed a novel feature extraction algorithm and a novel vehicle classification and recognition algorithm. Based on the features of the concavity and convexity areas, and the angles of concave and convex parts of the waveform, they developed an improved support vector machine (ISVM) classifier to perform vehicle classification and recognition. A wireless sensor network (WSN) for vehicle classification based on axle count and spacing was designed, calibrated, tested, and deployed near a weigh station in Sunol, California [3]. Bajwa et al. collected 53 different trucks exiting the weigh station and applied the Axle Detection algorithm (ADET) to successfully classify all 53 trucks. Kaewkamnerd et al. [5] studied vehicle classification system based the features of vehicle magnetic length, averaged energy, and Hill-pattern peaks. They classified vehicles into 4 types: motorcycle, car, pickup and van. Their system automatically detected vehicles, extracted features, and classified them with an average accuracy rate about 79.66%. Cheung and Varaiya [6] studied vehicle classification using magnetic sensors based on the 13-class Federal Highway Administration (FHWA) scheme. They studied three classifiers (k-Nearest Neighbor, Support Vector Machine and Direct Hill-Pattern Matching) in different datasets. Their average accuracy rate was about 70%. If the feature of magnetic length was considered, the average accuracy rate will be slightly improved. However, in order to obtain accurate magnetic length, more magnetic sensors are needed. The existing methods are either developed with too small sample size to verify the statistical robustness or lack of execution time analysis which cannot reveal the effectiveness for on-line vehicle classification in magnetic sensor nodes. Besides, some are vehicle-length-based or magnetic-length-based methods which use more than one sensor to implement vehicle classification. This will greatly increase the costs of system hardware and data processing.

The main objective of this paper is to develop a more effective vehicle classification method by a single multi-functional magnetic sensor for urban road. In this paper, a novel method is proposed to classify vehicles by measuring and processing field magnetic signal based on a single multi-functional sensor which is designed and developed. We firstly proposed the on-line vehicle classification method with a single multi-functional magnetic sensor and optimal MNS-based CART. The CART-based model had a simple structure and fast execution efficiency. And the multi-functional magnetic sensor was superior to the existing sensors in terms of reliability, cost, size and function. Furthermore, by comparison with some current methods, the proposed method was competitive in terms of accuracy rate, sample robustness and execution time. Finally, some experiments based on vehicle waveform data in a field road were implemented to test the performance of the new method.

In the rest of this paper, Section 2 describes the multi-functional magnetic sensor and data acquisition method

of this study. Section 3 discusses the steps of waveform feature extraction. Section 4 introduces the method proposed in this paper. Section 5 shows the experimental results. Finally, Section 6 concludes this paper.

2. Data acquisition

The wireless multi-functional magnetic sensor (referred to as 'sensor' in following texts) is used to acquire the vehicle waveform data. The sensor is designed and developed by our group (Fig. 1). Its shape parameters are diameter $D = 100$ mm and height $H = 64$ mm. The sensor is capable of detecting real-time magnetic field strength. Making use of the magnetoresistance effect, it translates the magnetic field strength into mV level voltage signal, and then into V level voltage signal by an amplifier. A filter is used to reduce environment noise. Then a digital output of the sensor will be obtained by an A/D converter. The values of digital output are corresponding to the variations that a vehicle disturbs the magnetic field. If there is no vehicle around the sensor, the digital output of the sensor would be an invariable value (eliminating the stochastic noises), which is defined as the benchmark value y_b . When there is a vehicle passing by the sensor, for the effect of ferromagnetic objects to magnetic field, the outputs of sensor are superposition values of the benchmark value y_b and the variations caused by the vehicle, which is defined as the vehicle waveform. The multi-functional sensor can detect several traffic flow parameters such as vehicle exist, vehicle stop, vehicle retrogradation, vehicle classification, traffic count, vehicle speed, occupancy ratios.

Due to the influences of environment and the sensor itself, the vehicle waveform data contain some high frequency noises (the sampling frequency $f_s = 333$ Hz in this paper). Therefore, a low-pass filter should be designed to eliminate high frequency noises. By analyzing the frequency distribution of noises, we designed a low-pass filter and obtained smooth vehicle waveforms.

With a low-pass filter, the vehicle waveform will be smoother and better for vehicle detection and feature extraction. A dual-window vehicle detection algorithm is used to extract the features of each vehicle waveform [16]. A vehicle waveform of car is shown in Fig. 2. y_b is the benchmark value. W_1 and W_2 are defined as the vehicle approach detection window (VADW) and the vehicle departure detection window (VDDW), respectively.

The dual-window vehicle detection algorithm is described as **Algorithm 1**.

Algorithm 1:

- 1: Initializing y_b and the size of W_1 and W_2
 - 2: Start W_1
 - 3: Input the vehicle waveform data in turn
 - if The data is located outside W_1
 - Note the sampling point of this data as o_1 , and go to step 4
 - else
 - Clear the data before the sampling point which is located inside W_1 , and go to step 2
 - end if
 - 4: Start W_2
 - 5: Input the vehicle waveform data in turn
 - if The data is located inside W_2
 - Note the sampling point of this data as o_2 , add the vehicle count by 1 and record the data between o_1 and o_2 as a vehicle waveform, then go to step 2 to detect next vehicle
 - else
 - Clear the data before the sampling point which is located outside W_2 , and go to step 4
 - end if
-

The integral waveform of each vehicle will be obtained using the dual-window vehicle detection algorithm. Fig. 3 shows some integral vehicle waveforms. For the differences of spatial distribution of ferromagnetic objects, the waveform features of each vehicle will be different, which is the foundation for vehicle classification.

3. Feature extraction

For a vehicle waveform, there are several features. The features which are easy to compute and useful to the work of vehicle classification should be extracted. For the same vehicle, when it passes the sensor at different speeds, the vehicle waveforms acquired at different time would not be the same with a fixed sampling frequency. For the time interval that a vehicle passes a sensor is small, the change of the vehicle speed can be ignored sometimes. Therefore, it can be supposed that a vehicle passes a sensor at a constant speed each time. Then, for the same vehicle, the



Fig. 1. Field traffic data acquisition by utilizing the wireless multi-functional magnetic sensor (a) wireless multi-functional magnetic sensor; (b) installation of the sensor; and (c) field traffic data acquisition on traffic road.

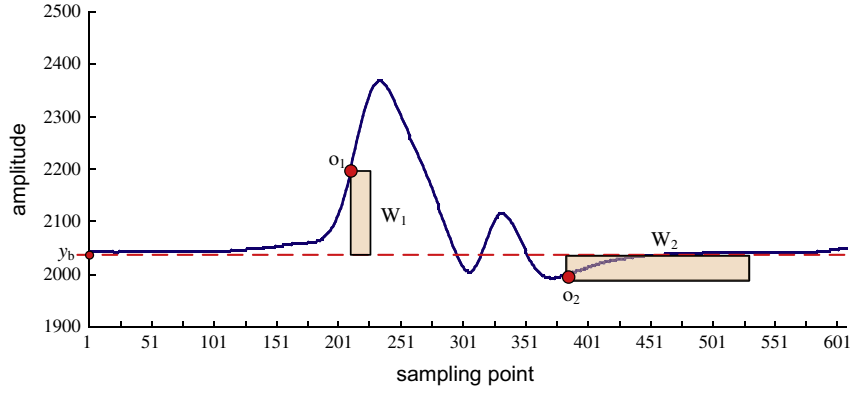


Fig. 2. Sketch map of dual-window vehicle detection algorithm.

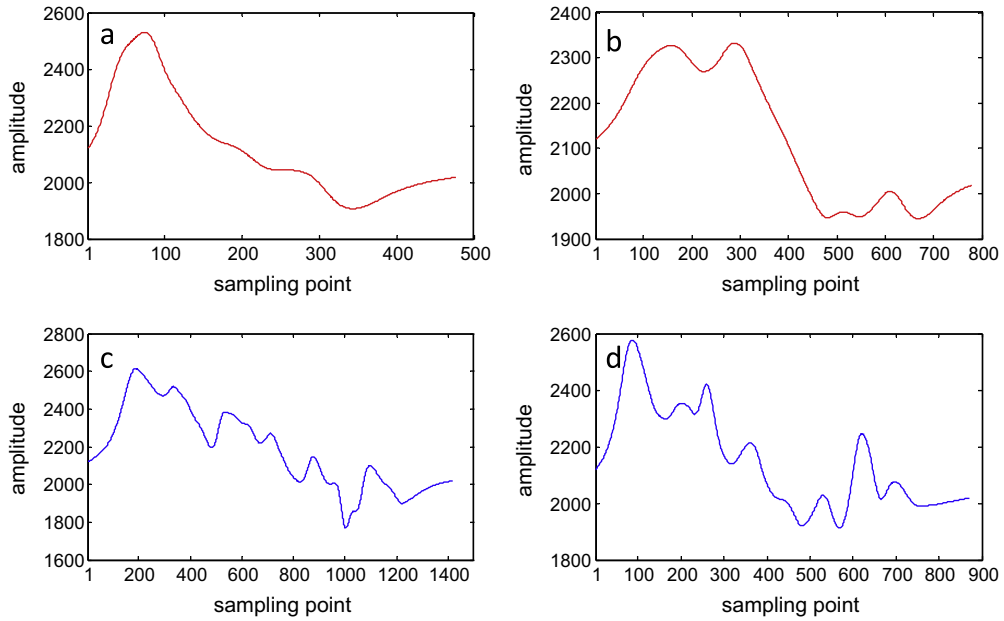


Fig. 3. Vehicle waveforms of cars and busses (a and b) cars; and (c and d) busses).

differences of the vehicle waveforms acquired at different times will be stretched or compressed at time axis. In order to eliminate the effect of speed on vehicle classification, some speed-independent features will be introduced. In this paper, 8 time-domain features are selected as the model inputs. Those features and their notations are the number of peaks (x_1), the maximum peak time ratio (x_2), the minimum trough time ratio (x_3), the mean value (x_4), the standard deviation (x_5), the maximum peak amplitude (x_6), the minimum trough amplitude (x_7) and the maximum peak/trough amplitude ratio (x_8). Denote vector $\mathbf{s}$ as a vehicle waveform series, that is $\mathbf{s} = [s_1, s_2, \dots, s_n]$, where $s_1, s_2, \dots, s_n$ are the sampling point of a vehicle waveform, and n is the number of the sampling points.

Fig. 4 shows the waveform of a vehicle and its time-domain features. o_1, o_2 are the start point and the end point of the vehicle waveform which are obtained by Algorithm

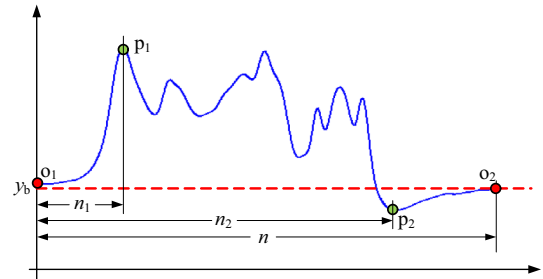


Fig. 4. A vehicle waveform and its time-domain features.

1; p_1, p_2 are the maximum peak point and minimum trough point; n_1, n_2 are the number of sampling points from o_1 to p_1 and o_1 to p_2 , respectively.

Denote n_p , n_t as the number of peaks and troughs, respectively, we have $|n_p - n_t| = 0$ or 1 . Then: $x_1 = n_p$, $x_2 = n_1/n$, $x_3 = n_2/n$, $x_4 = \text{ave}(\mathbf{s})$, $x_5 = \text{var}(\mathbf{s})$, $x_6 = \text{max}(\mathbf{s})$, $x_7 = \text{min}(\mathbf{s})$, $x_8 = \text{max}(\mathbf{s})/\text{min}(\mathbf{s})$, where ave , var , max , min are the mean function, the standard deviation function, the maximum function and the minimum function of vector $\mathbf{s}$, respectively.

Denote $\mathbf{x}$ as the feature vector of a vehicle waveform, then $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8]$. Features such as $x_1, x_2, x_3, x_6, x_7, x_8$ reflect the concentration and relative position of ferromagnetic material of a vehicle, which is the inherent property of the vehicle and therefore they are speed-independent. Features such as x_4, x_5 reflect the average distribution density and uniformity of ferromagnetic material. While different vehicle speeds lead to different numbers of sampling points, the average distribution density and uniformity of ferromagnetic material will be constant. Besides, the sensor outputs are uniform sampling based on both time and space. Therefore, the features x_4 and x_5 can be looked upon speed-independent ones.

Each vehicle waveform is replaced by 8 speed-independent features. Compared with traditional methods, these features will eliminate some errors caused by different vehicle speeds. When these 8 speed-independent features are selected as the model inputs, the time and space complexities of algorithm will be greatly reduced. So then, the algorithms of vehicle classification will be implemented easily in sensor nodes with limited memory and energy sources.

4. Methodology

4.1. Model and algorithm

Decision tree is a very efficient machine learning classification model and it uses binary tree to represent processing logic. The basic principle of a decision tree is that it recursively splits the data into subsets so that each subset contains a similar state or region of the data features. When this recursive process is done, a decision tree is created. Creating a decision tree is a process of tree growth that ceaselessly splits data set. Each split is corresponding to a decision problem, but also a node. For each split, a Max 'difference' between a subset and the rest of data set is required [17].

A decision tree is made of decision nodes, branches and leaves. The top node of decision nodes is the root node and each branch generates a new decision node or a leaf. Each decision node represents a problem or decision, and generally corresponds to a feature of an object to be classified. Each leaf node represents a possible classification result.

In this paper, the optimal MNS-based CART algorithm is used to create a decision tree. The CART algorithm adopts a binary recursive partitioning technology to split current sample set into two subsets, which makes the resulting decision tree has two branches for each non-leaf node. The decision tree which is based on CART algorithm is a binary tree with a simple structure [17]. The CART algorithm is provided in **Algorithm 2**:

Algorithm 2:

- 1: Create a root node R of sample set T
 - 2: **if** All the samples of T are the same class or T only contains τ samples
 then
 The node R is a leaf node, and return
 else
 Go to step 3
 end if
 - 3: For each feature x_i of T ($1 \leq i \leq N$, N is the number of features, in this paper $N = 8$), compute GINI values of different split cases, and denote the minimum GINI of x_i as $\text{GINI}(i)$
 - 4: **for** $i = 1 \rightarrow k$
 Find the minimum $\text{GINI}(i)$, select x_i as the decision feature of node R, and record the split case
 end for
 - 5: According to the split case in step 4, split sample set T into two subset T1 and T2
 - 6: Repeat step 1 to 5 for T1
 - 7: Repeat step 1 to 5 for T2
 - 8: A decision tree has been created
-

The algorithm parameter τ is the MNS which relates to the fitting degree of training samples based decision tree model. When $\tau = 1$, it will be the optimal division for the training samples, but it will result in an over-fitting decision tree. An over-fitting decision tree is not appropriate to classify new samples and will lead to low accuracy rates. Therefore, it is necessary to find an optimal τ for CART algorithm. Fortunately, the sample set is always limited and the model learning is an off-line process. We can find the optimal τ by computing the classification accuracy rates of all samples based on different decision trees trained with different values of τ . The value of τ corresponding to the maximum accuracy rate will be the optimal one.

Another algorithm parameter GINI is a measure of impurity for a sample set or subset [17], and GINI of sample set T is

$$\text{GINI}_T = 1 - \sum_{k=1}^c p(k)^2 \quad (1)$$

where k represents the k th class in set T, $1 \leq k \leq c$; c is the number of total classes; $p(k)$ is the (possibly weighted) relative frequency of class k in set T, and $\sum_{k=1}^c p(k) = 1$.

Assume that a split s of some feature divides set T into two subsets T1 and T2, and then the new GINI generated by s will be

$$\text{GINI}_T(s) = q \times \text{GINI}_{T1} + (1 - q) \times \text{GINI}_{T2} \quad (2)$$

where q is the (possibly weighted) fraction of samples going T1.

CART can achieve self-learning according to the input features, and can select the most effective features for growing a decision tree. CART uses a "priors" mechanism. Priors are akin to target class weights but they are invisible in that they do not affect any counts reported by CART in the tree. It automatically reweights the data to balance the classes, and ensures that the tree selected as optimal

minimizes balanced class error. The reweighting is implicit in the calculation of all probabilities and improvements and requires no user intervention [17]. To provide this flexibility, CART is unnecessary to those steps of eliminating invalid features and calculating feature correlation which are essential steps in other classification algorithms such as Artificial Neural Network or Naive Bayes.

4.2. Off-line training and optimal MNS τ

An optimal decision tree (T_o) which is able to effectively classify the known samples can be established by training those samples. Then the on-line classification process can be executed by making use of the T_o . Denote m as the number of training samples, we can obtain the feature matrix which is denoted as $\mathbf{X}_{m \times 8}$, that is $\mathbf{X}_{m \times 8} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots, \mathbf{x}_m]^T$. Where $\mathbf{x}_i$ is the feature vector of i th vehicle waveform and $\mathbf{x}_i = [x_{i1}, x_{i2}, x_{i3}, x_{i4}, x_{i5}, x_{i6}, x_{i7}, x_{i8}]$, $i = 1, 2, \dots, m$. Denote y_i as the corresponding vehicle class of i th vehicle waveform and the vehicle class vector will be $\mathbf{y} = [y_1, y_2, \dots, y_i, \dots, y_m]^T$.

By training sample feature matrix $\mathbf{X}_{m \times 8}$ and corresponding vehicle class vector $\mathbf{y}$ with different values of τ , a series of decision trees can be created. The decision tree which has a maximum classification accuracy rate of all samples will be selected as the T_o , and corresponding value of τ is the optimal τ . The structure of a decision tree can be described as following. A triangle represents a decision node which has two branches and a little dot represents a leaf node which is a vehicle class.

4.3. Pruning

Due to possible over-fitting for T_o , it will be necessary to create an optimal pruning tree PT_o by pruning a T_o sometimes. And then the PT_o will be more robust to new samples. CART algorithm uses afterwards pruning method. By pruning branches on an over-fitting tree, the nodes which are not pruned at the bottom will be denoted as leaf nodes. This paper adopts a MEP-based rule to prune T_o and selects a pruning tree with a maximum classification accuracy rate of the test samples as the optimal pruning tree PT_o .

Making use of the T_o and PT_o , it will be easy to implement a vehicle classification algorithm in a sensor node and to achieve on-line classification for urban traffic. In the trees of T_o and PT_o , the optimal features can be revealed. In order to reduce computation, calculating those optimal features will be enough for on-line vehicle classification in practice. On-line vehicle classification is an ergodic process along the decision tree from top to bottom. In the process, it will go to next branch based on the decision result of each decision node. The process will eventually reach a leaf node. Reading the information of the leaf node, a class of the new sample can be obtained.

5. Experiments

5.1. Vehicle samples

The magnetic sensors used for collecting vehicle waveform data in the paper are laid in the center of a lane of

Jiadadonglu road, Haidian district. From May, 2009 to November, 2009, we collected vehicle waveform data of 470 cars and 72 busses, and used the video to record the corresponding vehicle classes. Those 542 vehicle waveforms were selected as sample sets to test our method.

After two steps of vehicle waveform data acquisition and feature extraction, the feature matrix of 542 vehicle waveforms can be obtained (denoted as $\mathbf{X}_{542 \times 8}$). In order to train and test different methods, 271 vehicle waveform (including 235 cars and 36 busses) samples were selected randomly as the training samples in this paper (denoted as $\mathbf{X}_{\text{train}}$). And the rest were test samples (denoted as $\mathbf{X}_{\text{test}}$). At the same time, we used the videos to count the actual vehicle classes. Denote 1 as the car, 2 as the bus, and then we got the vehicle class vector $\mathbf{y}_{\text{train}}$ corresponding to $\mathbf{X}_{\text{train}}$ and vehicle class vector $\mathbf{y}_{\text{test}}$ corresponding to $\mathbf{X}_{\text{test}}$.

5.2. Optimal MNS τ

In order to obtain the optimal τ , accuracy rates of samples $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ which are based on different decision trees with different values of τ were computed (Table 1). As shown in Table 1, when τ is small, the classification accuracy rate of $\mathbf{X}_{\text{train}}$ will be 100%, but the accuracy rate and adaptability of $\mathbf{X}_{\text{test}}$ are poor. As τ increases, the accuracy rate of $\mathbf{X}_{\text{train}}$ will fall; however, the accuracy rate of $\mathbf{X}_{\text{test}}$ can be improved. When τ is big enough, the decision tree will not be able to classify the samples. In principle, τ should not be larger than the number of a class of least samples. Otherwise, it will make against the classification of a class of samples with least number. In this paper, the averages of two accuracy rates of $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ will be computed with different τ , and the corresponding value of τ with maximum average rate will be the optimal one.

By computing the average classification accuracy rates of the sample set $\mathbf{X}_{542 \times 8}$, Fig. 5 shows the average accuracy rates of $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ which are based on decision trees with different values of τ . It can be drawn that when τ is 4, 5, 6, 7, 8 or 9, we have the maximum average accuracy rate. The greater the value of τ is, the faster the speed of algorithm converge will be, therefore, the optimal τ is 9 in this paper.

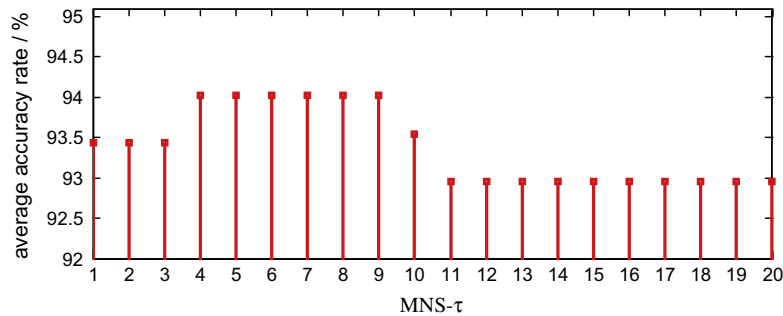
5.3. Results

When $\tau = 9$, through training the samples of $\mathbf{X}_{\text{train}}$ and $\mathbf{y}_{\text{train}}$, the optimal decision tree T_o can be obtained based on CART (Fig. 6). In Fig. 6, the triangles represent decision nodes, the dots represent leaf nodes, and 1 represents that the result of vehicle classification is a car, 2 represents a bus.

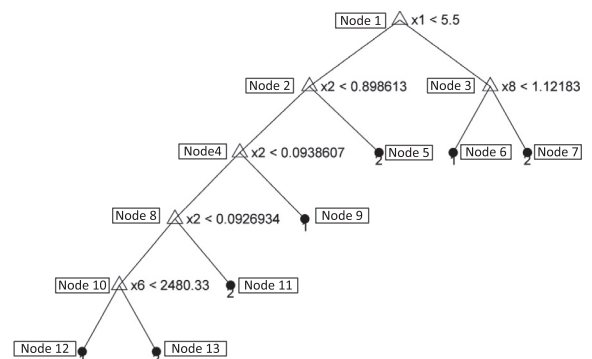
From T_o , we have the best combination of waveform features $\mathbf{x}_{\text{opt}} = [x_1, x_2, x_6, x_8]$. Using the decision tree T_o to implement on-line vehicle classification, we just need to calculate the four features x_1 , x_2 , x_6 and x_8 . Derived from Fig. 6, the on-line classification algorithm (Algorithm 3) can be shown as follows:

Table 1Accuracy rates of samples $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ based on different decision trees with different τ .

τ	Accuracy rate of $\mathbf{X}_{\text{train}}$			Accuracy rate of $\mathbf{X}_{\text{test}}$		
	Bus	Car	Average	Bus	Car	Average
1	1.000	1.000	1.000	0.750	0.987	0.869
2	1.000	1.000	1.000	0.750	0.987	0.869
3	1.000	1.000	1.000	0.750	0.987	0.869
4	1.000	0.996	0.998	0.778	0.987	0.883
5	1.000	0.996	0.998	0.778	0.987	0.883
6	1.000	0.996	0.998	0.778	0.987	0.883
7	1.000	0.996	0.998	0.778	0.987	0.883
8	1.000	0.996	0.998	0.778	0.987	0.883
9	1.000	0.996	0.998	0.778	0.987	0.883
10	0.972	0.996	0.984	0.778	0.996	0.887
11	0.944	0.996	0.970	0.778	1.000	0.889
12	0.944	0.996	0.970	0.778	1.000	0.889
13	0.944	0.996	0.970	0.778	1.000	0.889
14	0.944	0.996	0.970	0.778	1.000	0.889
15	0.944	0.996	0.970	0.778	1.000	0.889
16	0.944	0.996	0.970	0.778	1.000	0.889
17	0.944	0.996	0.970	0.778	1.000	0.889
18	0.944	0.996	0.970	0.778	1.000	0.889
19	0.944	0.996	0.970	0.778	1.000	0.889
20	0.944	0.996	0.970	0.778	1.000	0.889
36 ^a	0.944	0.991	0.968	0.778	1.000	0.889
235 ^b	0.944	0.991	0.968	0.778	1.000	0.889
271 ^c	0.917	0.991	0.954	0.778	1.000	0.889
>271	0.000	1.000	0.500	0.000	1.000	0.500

^a Number of busses in $\mathbf{X}_{\text{train}}$.^b Number of cars in $\mathbf{X}_{\text{train}}$.^c Number of cars and busses in $\mathbf{X}_{\text{train}}$.**Fig. 5.** Average accuracy rates of training and test samples based on decision trees with different τ .**Algorithm 3:**

- 1: node 1 if $x_1 < 5.5$ then node 2 else node 3
- 2: node 2 if $x_2 < 0.898613$ then node 4 else node 5
- 3: node 3 if $x_8 < 1.12183$ then node 6 else node 7
- 4: node 4 if $x_2 < 0.0938607$ then node 8 else node 9
- 5: node 5 class = 2
- 6: node 6 class = 1
- 7: node 7 class = 2
- 8: node 8 if $x_2 < 0.0926934$ then node 10 else node 11
- 9: node 9 class = 1
- 10: node 10 if $x_6 < 2480.33$ then node 12 else node 13
- 11: node 11 class = 2
- 12: node 12 class = 1
- 13: node 13 class = 2

**Fig. 6.** Optimal decision tree based on CART by training the training samples.

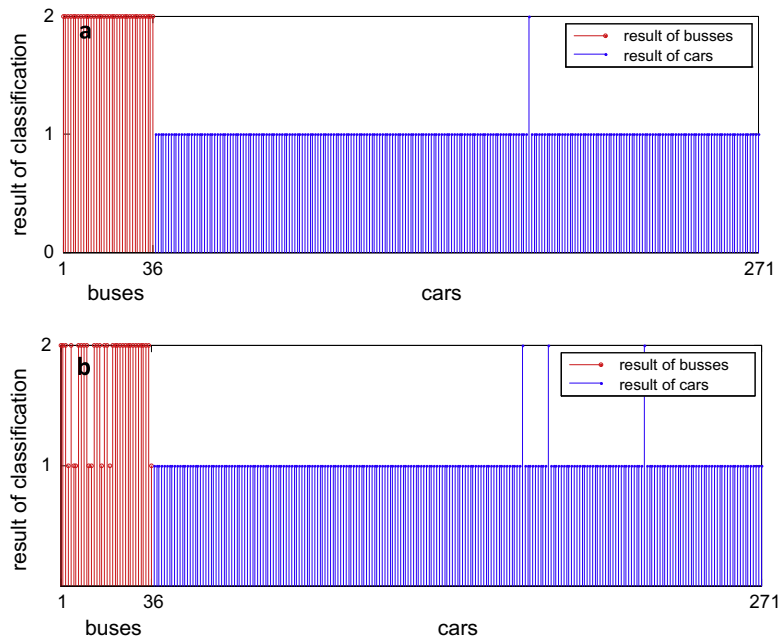


Fig. 7. Distribution graphs of vehicle classification based on T_o (a) training samples $\mathbf{X}_{\text{train}}$; and (b) test samples $\mathbf{X}_{\text{test}}$.

Table 2

Results of training samples $\mathbf{X}_{\text{train}}$ based on T_o .

	Cars (tree output)	Busses (tree output)	Accuracy rate (%)
Cars (235)	234	1	99.6
Busses (36)	0	36	100.0
Average			99.8

Table 3

Results of test samples $\mathbf{X}_{\text{test}}$ based on T_o .

	Cars (tree output)	Busses (tree output)	Accuracy rate (%)
Cars (235)	232	3	98.7
Busses (36)	8	28	77.8
Average			88.3

By using **Algorithm 3**, we respectively take the sets of training samples $\mathbf{X}_{\text{train}}$ and test samples $\mathbf{X}_{\text{test}}$ to test the validity and effectiveness of our method. Test results of $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ are shown in Fig. 7.

By counting the test results in Fig. 7, the accuracy rates of $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ are shown in Tables 2 and 3.

In order to improve the classification accuracy of the test samples, we obtain the optimal pruning tree PT_o by pruning the T_o based on the MEP rule. As shown in Fig. 8, the dotted lines are some parts of the T_o and the rest are the PT_o .

Based on PT_o in Fig. 8, a new on-line classification algorithm is shown in **Algorithm 4**.

Algorithm 4:

- 1: node 1 if $x_1 < 5.5$ then node 2 else node 3
- 2: node 2 if $x_2 < 0.898613$ then node 4 else node 5
- 3: node 3 if $x_8 < 1.12183$ then node 6 else node 7
- 4: node 4 class = 1
- 5: node 5 class = 2
- 6: node 6 class = 1
- 7: node 7 class = 2

Compared with **Algorithm 3**, **Algorithm 4** is more concise and its computation will be lower, which is very important to implement on-line vehicle classification in a sensor node with very limited arithmetic and memory capacity. To adopt MEP-based rule to execute pruning operation of T_o , **Algorithm 4** has a higher classification accuracy and robustness for new samples than **Algorithm 3**. Based on **Algorithm 4**, test results of $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ are shown in Tables 4 and 5 respectively. As we can see, the accuracy rate of $\mathbf{X}_{\text{train}}$ based on PT_o has been a little bit declined, but the accuracy rate of $\mathbf{X}_{\text{test}}$ will be improved. To improve the accuracy of new samples is quite important to on-line vehicle classification on the field road.

5.4. Comparison and discussion

In order to make a further verification of the effectiveness of the proposed method, we compared the method with a variety of current classification methods in accuracy rate, sample robustness and execution time. Based on $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$, Table 6 presented the results of the classification accuracy rates of CART algorithm and other current classification methods such as Template Matching (TPM), Artificial Neural Network (ANN), Support Vector Machine

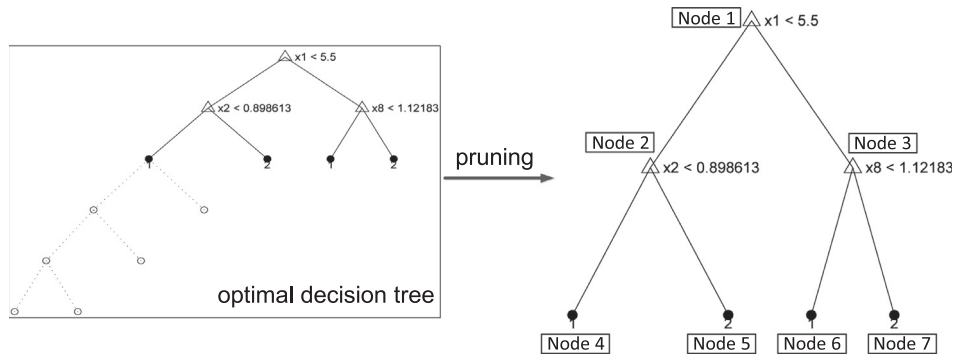


Fig. 8. Optimal pruning tree PT_o by pruning T_o based on MEP.

Table 4

Results of training samples $\mathbf{X}_{\text{train}}$ based on PT_o .

	Cars (tree output)	Busses (tree output)	Accuracy rate (%)
Cars (235)	234	1	99.6
Busses (36)	2	34	94.4
Average			97.0

Table 5

Results of test samples $\mathbf{X}_{\text{test}}$ based on PT_o .

	Cars (tree output)	Busses (tree output)	Accuracy rate (%)
Cars (235)	235	0	100.0
Busses (36)	8	28	77.8
Average			88.9

Table 6

Accuracy rates of different methods by original samples of $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$.

Methods	Object	Action	Accuracy rate of car (%)	Accuracy rate of bus (%)	Average (%)
TPM	$\mathbf{X}_{\text{train}}$	Training	92.3	72.2	82.3
	$\mathbf{X}_{\text{test}}$	Testing	79.1	19.4	49.3
ANN	$\mathbf{X}_{\text{train}}$	Training	98.7	83.3	91.0
	$\mathbf{X}_{\text{test}}$	Testing	100.0	69.4	84.7
SVM	$\mathbf{X}_{\text{train}}$	Training	99.6	86.1	92.8
	$\mathbf{X}_{\text{test}}$	Testing	100.0	77.8	88.9
NBayes	$\mathbf{X}_{\text{train}}$	Training	99.1	66.7	82.9
	$\mathbf{X}_{\text{test}}$	Testing	99.6	66.7	83.1
CART	$\mathbf{X}_{\text{train}}$	Training	99.6	94.4	97.0
	$\mathbf{X}_{\text{test}}$	Testing	100.0	77.8	88.9

Table 7

Accuracy rates of different methods by swapping samples of $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$.

Methods	Object	Action	Accuracy rate of car (%)	Accuracy rate of bus (%)	Average (%)
TPM	$\mathbf{X}_{\text{test}}$	Training	91.5	55.6	73.6
	$\mathbf{X}_{\text{train}}$	Testing	100.0	0.0	50.0
ANN	$\mathbf{X}_{\text{test}}$	Training	99.6	77.8	88.7
	$\mathbf{X}_{\text{train}}$	Testing	99.6	80.6	90.2
SVM	$\mathbf{X}_{\text{test}}$	Training	100.0	80.6	90.3
	$\mathbf{X}_{\text{train}}$	Testing	100.0	77.8	88.9
NBayes	$\mathbf{X}_{\text{test}}$	Training	99.6	61.1	80.4
	$\mathbf{X}_{\text{train}}$	Testing	99.1	63.9	81.5
CART	$\mathbf{X}_{\text{test}}$	Training	99.6	97.2	98.4
	$\mathbf{X}_{\text{train}}$	Testing	97.0	91.7	94.4

(SVM), Naive Bayes classification (NBayes). In Table 6, the CART algorithm has a higher accuracy both in tests of $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$. Because of the smaller sample size of busses, the training of those algorithms will be not enough. This will result in a low accuracy rate of bus samples for those methods, but CART and SVM can present better results. In addition, in the learning process of samples, CART algorithm can get a stable classifier model with a higher fitting degree (See results of $\mathbf{X}_{\text{train}}$ in Table 6).

To test the sample robustness of CART algorithm, we swapped $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ in the next experiment. Then, for each classification method, the particularity of results caused by the particularity of samples could be avoided. Table 7 presented the accuracy rates different methods by swapping $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$. Compared with Table 6, CART algorithm still can obtain a higher classification accuracy

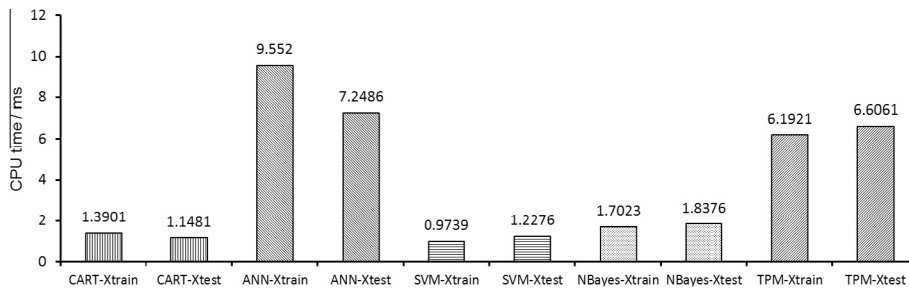


Fig. 9. CPU time of different methods to classify samples of X_{train} and X_{test} .

rate for different samples, and then SVM followed. Therefore, conclusions can be drawn from Tables 6 and 7 that CART algorithm can obtain higher classification accuracy in those methods and its sample robustness is better, too.

Due to the off-line processes of algorithm learning and model optimization proposed in this paper, there is no need to consider the timeliness. However, using the trained model to implement vehicle classification is an on-line process in sensor nodes whose energy and computing resources are limited. The process is an on-line real-time classification problem. For execution time, we mainly focus on the time of on-line vehicle classification. Fig. 9 shows the computer CPU time of the different methods to classify samples of X_{train} and X_{test} (the results are the average values of test results in ten times).

In Fig. 9, ANN and TPM are not suitable for on-line classification due to their longer CPU time. In terms of on-line classification speed, CART, SVM and NBayes are better and their CPU time is small. But considering the accuracy rate of classification, the advantage of CART is more obvious, the second is SVM.

6. Conclusions

Using self-developed multi-functional magnetic sensors, the vehicle waveforms can be acquired in real time. The magnetic sensor transmits data by radio, and it is convenient to deploy in urban roads. Besides, the sensor has a low cost in hardware, maintenance and implementation. Through two steps of vehicle waveform data processing and feature extraction, the vehicle waveform features which are suitable to vehicle classification for urban road can be obtained. Making use of those waveform features, the MNS-based CART proposed in this paper enables efficient vehicle classification. By pruning the optimal decision tree based on MEP rule, the new optimal pruning tree can effectively improve the accuracy and robustness for new samples. For those do not need different features' correlation analysis, this method can give an optimal feature combination by automatically filtering and self-learning. Compared with several current classification methods, the proposed method has obvious advantages in terms of accuracy rate, sample robustness and execution time.

Moreover, comparing with traditional classification methods, the proposed method does not require vehicle or magnetic length information, and achieves on-line

vehicle classification by just a single multi-functional magnetic sensor. Therefore, this method also has advantages in terms of cost and complexity of application system realization. In addition, according to the multifarious application requirements for vehicle classification, we can train various decision tree models to meet different applications of vehicle classification, which need to collect more samples of different vehicle classes, and it is one of main directions for future researches.

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